1. Student Details :

- **Name:** Vikas
- **Roll Number:** 23f3001800
- **Course:** Application Development II (MAD-2), Jan 2026
- **About me**: Exploring the world of data science and AI.
- **Presentation video** : project presentation

Project Title : CampusHire — From Classroom to Career

Problem Statement :

Institutes require efficient systems to manage campus recruitment activities involving companies and students. Currently, many institutes rely on spreadsheets, emails, or manual coordination, which makes it difficult to manage company approvals, placement drives, student registrations, and application tracking.

Approach :

I built CampusHire as a modern decoupled application. The backend strictly serves data via REST APIs, while the frontend handles all the UI routing and state management, resulting in a fast and seamless user experience.

- **Backend :** Built an API-first backend using Flask and Flask-RESTful. It handles token-based authentication via Flask-Security-Too and manages the core business logic, like validating if a student meets the criteria before allowing an application to a drive.
- **Frontend :** Developed a Single Page Application (SPA) using Vue 3 and Pinia. It dynamically renders dashboards based on the user's role (Admin, Company, or Student) and uses Axios to fetch data from the backend APIs asynchronously.
- **Background Jobs :** Integrated Celery with Redis as the message broker. This handles heavy lifting outside the main request cycle, specifically: sending daily deadline reminders to students, generating the monthly HTML activity report for the Admin, and processing user-triggered CSV application exports.
- **Performance Optimization :** Implemented Redis caching for frequently accessed, read-heavy API endpoints to reduce database load. Background jobs ensure the UI never freezes while waiting for email servers or file generation.

3. AI / LLM Usage Declaration :

AI was primarily used to assist with the structural boilerplate of the API and Vue components, as well as debugging complex asynchronous tasks

Component / Tool	Function	Usage	Notes
Vue.js (Core + Router)	Reactive SPA frontend development, routing.	18%	Central SPA framework.
Flask (App + API)	Backend logic, serving REST APIs.	18%	Exclusively for REST API.
SQLite + SQLAlchemy	Data persistence, relational ORM.	12%	Standard ORM implementation.
Auth (Security-Tool)	Login/logout, token-based role access control.	10%	Focus on token security.
Bootstrap / CSS	Styling, layout, responsive UI.	25%	Used for responsive design.
User Logic (Applications)	Core business flow, application management.	7%	Business logic via frontend/API.
Axios / Fetch	Managing asynchronous API calls/data fetching.	12%	Facilitates async state/data retrieval.
Admin CRUD Logic	Frontend interfaces for user/drive/company CRUD.	5%	Administrative management tools.
APIs / Integration	Structuring REST endpoint data communication.	10%	Focus on Blueprint/resource structuring.
HTML + Jinja2	Minimal server-side template rendering for entry point.	25%	Only for main entry page.

Database Design (~ 50 words) :

The schema utilizes a relational SQLite database via SQLAlchemy. It anchors on a unified `User` table for authentication, extending into distinct `Student` and `Company` profiles. An

`Application` junction table securely maps students to company-created `PlacementDrives`, enabling real-time status tracking and maintaining comprehensive historical placement records.

What I implemented :

I built a complete, role-based placement portal. Admins can govern the platform and view stats; Companies can post drives and manage applicant pipelines (including scheduling interviews and uploading offer letters); Students can track their job hunts, upload resumes, and view their placement history. I also integrated Redis and Celery to sync CSV exports and scheduled email reports.

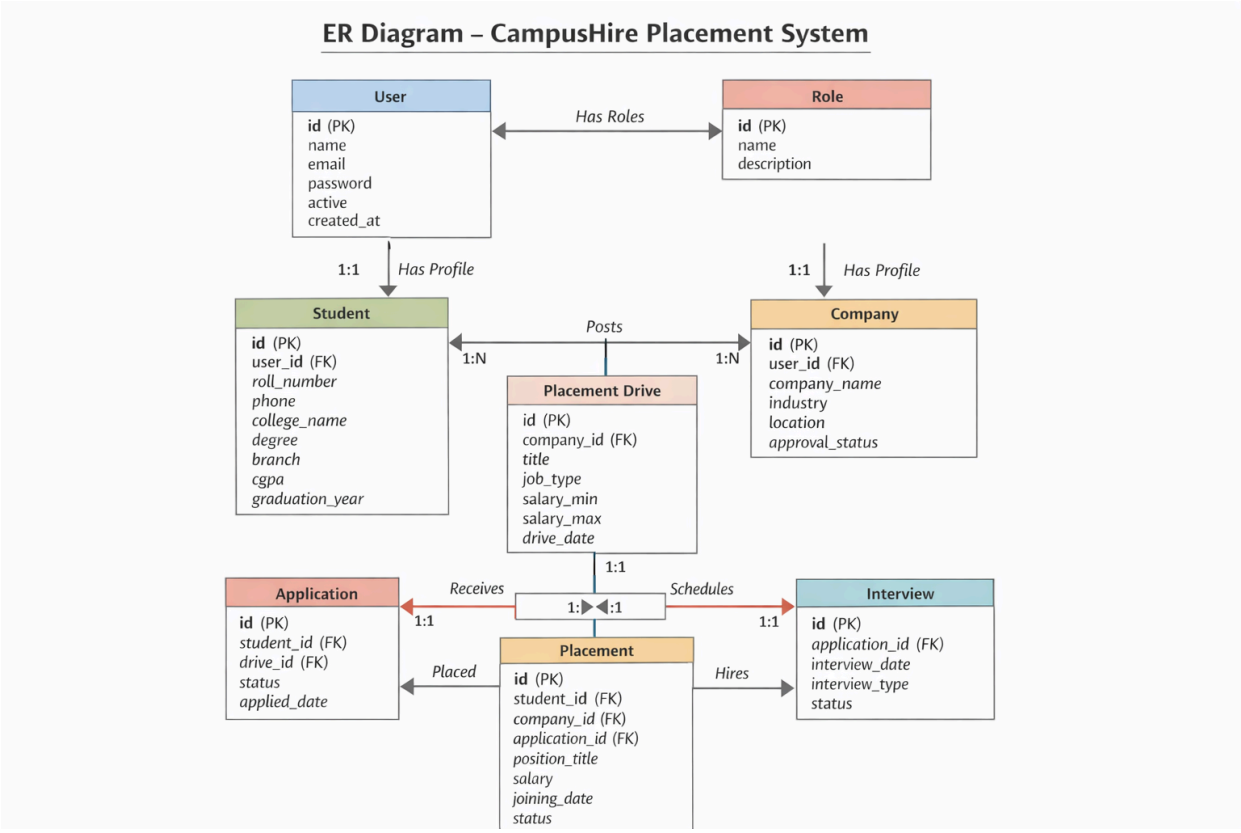
4. Technologies and Frameworks :

- **Backend:** Flask 3.x, Flask-RESTful
- **Authentication:** Flask-Security-Too (Token-based)
- **Frontend:** Vue 3 (Composition API), Pinia, Vue Router, Axios
- **Database:** SQLite, SQLAlchemy ORM
- **Async & Caching:** Redis, Celery 5.x, Flask-Caching
- **Styling:** Bootstrap 5

5. Database Schema :

- **User:** Handles authentication and role identification (`email`, `password`, `active`).
- **Student:** Extends User; stores academic details (CGPA, branch, resume file path).
- **Company:** Extends User; stores organization details (name, HR contact).
- **PlacementDrive:** Created by Companies, requires Admin approval. Stores job descriptions, eligibility criteria, and deadlines.
- **Application:** Tracks the many-to-many relationship between Students and Drives, including current progression status (Applied, Shortlisted, Selected)

ER Diagram :



6. API Resource Endpoints :

Note: All endpoints are prefixed with `/api`.

Role	Method	Path	Description
Student	GET/PUT	`/student/<id>`	Fetch/update profile
Student	GET	`/student/<id>/applications`	View applied drives
Student	POST	`/student/<id>/apply/<drive_id>`	Apply to drive
Student	DELETE	`/student/<id>/applications/<app_id>`	Withdraw application
Student	POST	`/student/<id>/ex`	Export CSV history

		`port/csv`	
Company	POST/PUT	`/company/<id>`	Create/update profile
Company	GET/POST	`/company/<company_id>/drives`	List/create drives
Company	GET	`/company/<company_id>/drives/<drive_id>/applicants`	View drive applicants
Company	POST	`/company/<company_id>/applications/<app_id>/interview`	Schedule interview
Company	PATCH	`/company/<company_id>/applications/<app_id>/selection`	Update selection status
Admin	GET	`/admin/stats`	Fetch dashboard stats
Admin	GET	`/admin/search`	Global search
Admin	GET	`/admin/applications`	View all applications
Shared	GET	`/drives`	Fetch approved drives
Shared	GET	`/uploads/resumes/<filename>`	Serve user resumes
Shared	POST	`/upload-offer`	Upload offer letters

System Architecture :

CampusHire utilizes a decoupled Client-Server architecture. The Vue 3 client runs in the browser, requesting and submitting JSON data via Axios to the Flask REST API. The API reads and writes to a SQLite database. To ensure the UI remains responsive, heavy tasks (like CSV generation and bulk emails) are passed to a Redis message broker and executed asynchronously by Celery workers.

Core Features Implemented :

- Multi-role Token-based authentication and restricted dashboards (Admin, Student, Company).
- Admin approval workflows for new company registrations and placement drives.
- Automated drive eligibility filtering for students based on CGPA and branch.
- File handling for student resumes and company offer letters.
- Celery-powered scheduled jobs (Daily deadline reminders, Monthly placement reports).
- All User-triggered synchronous CSV data export for student application history, drives and company students, and placements.
- An offer letter is generated by the company and students can see the offer letter and accept and reject the offer letter.
-

9. Learning Outcomes :

A major learning outcome was building strict REST APIs to serve a separate frontend application. Managing global state with Pinia taught me how to keep the frontend UI fully in sync without forcing the user to refresh the page. Setting up Celery with Redis also provided highly practical experience in handling long-running background processes so the main application stays fast, responsive, and professional.